| Reg. No. | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

# B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, NOVEMBER 2025

Second Semester

## 21ECC101J - ELECTRONIC SYSTEM AND PCB DESIGN

*(For the candidates admitted during the academic year 2021 - 2022 to 2024 - 2025)*

**Note:**

i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
ii. **Part - B** and **Part - C** should be answered in answer booklet.

**Time: 3 hours** **Max. Marks: 75**

**PART - A (20 × 1 = 20 Marks)**
Answer **all** Questions

| Question | Marks | CO | BL |
|---|---|---|---|
| 1. Mass action law for semiconductors in electronic devices is ........................<br>(A) $n \times p = ni^2$ (B) $n \times p = ni$<br>(C) $n \times p = ni3$ (D) $n \times p = ni1/2$ | 1 | 1 | 1 |
| 2. Which of the following is current amplification factor for CE configuration........<br>(A) $\alpha = IB/IC$ (B) $\beta = IC/IB$<br>(C) $\gamma = IE/IB$ (D) $z = IV/IB$ | 1 | 1 | 1 |
| 3. What type of material is obtained when an intrinsic semiconductor is doped with pentavalent impurity?<br>(A) Extrinsic semiconductor (B) Insulator<br>(C) N-type semiconductor (D) P-type semiconductor | 1 | 1 | 1 |
| 4. In which region does BJT act as the ON switch in electronic circuits?<br>(A) Cut-off (B) Saturation<br>(C) Active (D) Reverse saturation | 1 | 1 | 1 |
| 5. Schottky diodes are primarily used in<br>(A) High-power rectifiers (B) High-speed switching applications<br>(C) LED circuits (D) Low-frequency filters | 1 | 2 | 1 |
| 6. Power BJT is primarily used for<br>(A) Low-power signal amplification (B) Power switching and amplification<br>(C) High-speed switching circuits (D) Voltage regulation in ICs | 1 | 2 | 1 |
| 7. An SCR has how many layers and junctions?<br>(A) 2 layers, 1 junction (B) 4 layers, 3 junctions<br>(C) 3 layers, 2 junctions (D) 5 layers, 4 junctions | 1 | 2 | 1 |
| 8. In a Gunn diode, the negative resistance characteristic is due to<br>(A) Impact ionization (B) Carrier recombination<br>(C) Junction capacitance (D) Transfer of electrons from lower valley to higher energy valley | 1 | 2 | 2 |

9. What is the main function of a rectifier? 1 3 1
   (A) To amplify AC signals
   (B) To store electrical energy
   (C) To convert AC to DC
   (D) To increase voltage

10. Which instrument is used to measure electric current in a circuit? 1 3 1
   (A) Digital voltmeter
   (B) Digital multimeter
   (C) Ohmmeter
   (D) Ammeter

11. Clipper is used as a 1 3 1
   (A) Limiter
   (B) regulator
   (C) Filter
   (D) Capacitor

12. What is the maximum efficiency of a half-wave rectifier? 1 3 2
   (A) 40.6%
   (B) 50%
   (C) 81.2%
   (D) 100%

13. Which of the following is the first step in PCB layout planning? 1 4 1
   (A) Routing traces
   (B) Component placement
   (C) Placing vias
   (D) Designing power planes

14. Which of the following is NOT a common type of PCB? 1 4 1
   (A) Quad-layer
   (B) Single-layer
   (C) Double-layer
   (D) Multi-layer

15. What is the purpose of a ground fill (or copper pour) in PCB design? 1 4 1
   (A) Reduce the cost of copper
   (B) Enhance aesthetic appeal
   (C) Improve thermal and EMI performance
   (D) Minimize board thickness

16. What is the main purpose of Design Rule Check (DRC) in PCB design? 1 4 2
   (A) To detect layout violations like spacing and width
   (B) To ensure PCB fits in enclosure
   (C) To reduce manufacturing cost
   (D) To verify component values

17. Crosstalk in PCB design is caused by: 1 5 2
   (A) Power supply voltage fluctuations
   (B) Overlapping ground planes
   (C) Electromagnetic coupling between adjacent traces
   (D) Improper soldering

18. Which of the following is a result of poor power distribution network (PDN) design in digital PCBs? 1 5 1
   (A) Voltage noise and instability
   (B) Overheating
   (C) Signal distortion
   (D) Crosstalk

19. Which rule is crucial when routing differential pair signals? 1 5 2
   (A) Route one on top and the other on bottom layer
   (B) Use the thickest possible traces
   (C) Keep the pair as short as possible
   (D) Match the trace lengths

| | | Marks | CO | BL |
|---|---|---|---|---|
| 20. | Why should analog and digital components be separated during layout planning?<br>(A) To reduce board size (B) To minimize interference and noise<br>(C) To improve mechanical strength (D) To avoid soldering errors | 1 | 5 | 1 |

## PART - B (5 × 8 = 40 Marks)

Answer **all** Questions

| | | Marks | CO | BL |
|---|---|---|---|---|
| 21. a) | Define NPN transistor. Draw and explain the input and output characteristics of common emitter configuration. | 8 | 1 | 2 |
| | **(OR)** | | | |
| b) | Define JFET. Explain the operation of n-channel JFET | 8 | 1 | 2 |
| 22. a) | Explain the working of a Gunn diode. Draw characteristic curves and explain its negative resistance property. Also, discuss its application in microwave circuits | 8 | 2 | 1 |
| | **(OR)** | | | |
| b) | Compare and contrast the operating principles and applications of Schottky Diode and IMPATT Diode. | 8 | 2 | 1 |
| 23. a) | Discuss the principle and operation of main components of Cathode Ray Oscilloscope (CRO). | 8 | 2 | 2 |
| | **(OR)** | | | |
| b) | Draw the block diagram of power supply and explain the requirement of each block | 8 | 2 | 2 |
| 24. a) | Discuss the concept of PCB design and its significance in electronic system. What are the rules for component placements on PCB | 8 | 4 | 1 |
| | **(OR)** | | | |
| b) | Draw the block diagram of layout planning for PCB design and give the explanation of each block | 8 | 4 | 1 |
| 25. a) | Describe the design rules for PCB for microwave circuits. | 8 | 5 | 1 |
| | **(OR)** | | | |
| b) | Discuss the impact of environmental factors such as humidity, temperature, chemical exposure and vibration on PCB design. | 8 | 5 | 1 |

## PART - C (1 × 15 = 15 Marks)

Answer **any 1** Question

| | | Marks | CO | BL |
|---|---|---|---|---|
| 26. | Discuss the working of Full Wave Rectifier with necessary diagrams. Find out Vdc, Idc, Vr.m.s, Ir.m.s, efficiency and TUF for Full wave Rectifier circuit. | 15 | 3 | 4 |
| 27. | Explain all the steps involved in the monolithic fabrication process of integrated circuits. | 15 | 5 | 1 |

* * * * *